

Week 1 — Business Understanding Report

Field	Detail
Programme	Machine Learning Internship
Week	Week 1 — Machine Learning Problem Framing & Data Understanding
Project	Customer Churn Prediction Project
Client	ABC Communications Ltd
Role	Junior Machine Learning Engineer, AnalystLab Africa Consulting
Dataset	Telco Customer Churn (Kaggle — IBM Sample Data Sets)

1. Business Context

ABC Communications Ltd, a client of AnalystLab Africa Consulting, wishes to predict customer cancellation ("churn") before it happens. The objective of this first week is to understand the business problem, inspect the dataset, and prepare for the model development planned for the following weeks.

2. Preliminary Research — Key Definitions

- **Machine Learning:** A set of algorithms that learn patterns from historical data to make predictions, without all rules being explicitly programmed.
- **Predictive Analytics:** The use of historical data combined with statistical or Machine Learning methods to estimate the likelihood of a future event occurring.
- **Customer Churn:** When a customer stops using a service (cancellation, leaving for a competitor). In the telecom sector, this corresponds to the loss of a line or a contract.

3. Answers to Business Questions

3.1 What business problem must the model solve?

ABC Communications wants to identify, in advance, customers with a high probability of churning. It costs significantly more to acquire a new customer than to retain an existing one: by knowing at-risk customers beforehand, the company can take targeted actions (loyalty offers, customer service calls, commercial discounts) rather than reacting after the customer has left.

3.2 What is the target variable?

The Churn column in the dataset: a binary value (Yes / No) indicating whether the customer cancelled within the last month.

3.3 Classification or regression?

Since the target is a binary category (Yes/No) and not a continuous numerical value, this is a binary classification problem.

3.4 What evaluation metrics should be used?

Churn is an imbalanced problem (many more "No" than "Yes"): accuracy alone would be misleading; a model that always predicts "No" would already show good accuracy without being useful. It is better to track:

- **Recall (rappel):** Among the customers who actually leave, how many are detected by the model? Essential because missing an at-risk customer is costly.
- **Precision:** Among the customers predicted as "at risk", how many actually are? Essential because retention actions have a cost.
- **F1-score:** The balance between precision and recall.
- **ROC-AUC:** The overall ability of the model to distinguish between the two classes.

3.5 Which algorithms are suitable?

For a binary classification on mixed tabular data (numerical + categorical), the shortlisted candidates are:

- **Logistic Regression** — A good starting point, simple and interpretable; useful for explaining churn factors to the client.
- **Decision Tree / Random Forest** — Captures non-linear interactions, provides feature importance.
- **Gradient Boosting (XGBoost / LightGBM)** — Generally the best performing on this type of data.
- **KNN / SVM** — Worth considering for comparison, but less common in production for this type of use case.

Recommendation: Start with Logistic Regression and Random Forest, then compare their performances, as the client likely wants to understand the churn factors and not just get a "black box" prediction.